



ITAI 2373 · FINAL PROJECT

NewsBot Intelligence System 2.0

An advanced NLP platform for classification, topic discovery, summarization, multilingual analysis, and conversational news intelligence.

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ITAI 2373 — Natural Language Processing | Houston Community College

What We'll Cover

01

Project Overview

Motivation, goals, and the four-module architecture

02

Content Analysis Engine

Classification, topic modeling, sentiment, entity mapping

03

Language & Multilingual AI

Summarization, semantic search, translation, cultural context

04

Conversational Interface

Natural language queries and intent understanding

05

Integration, Testing & Results

System evaluation, metrics, and business impact

4 Integrated Modules

A

Content Analysis

B

Language & Generation

C

Multilingual Intelligence

D

Conversational Interface

WHY IT MATTERS

News Moves Fast. Understanding It Shouldn't Be Slow.

Newsrooms, analysts, and researchers are flooded with articles across topics, languages, and sources — with no easy way to detect trends, sentiment shifts, or hidden connections at scale.

1M+

Articles published daily
worldwide

7+

Languages a global newsroom
must track

Hours

Lost to manual reading & tagging

NewsBot 2.0 solves this: one system that classifies, summarizes, translates, and converses about the news — automatically.



Tech layoffs spark market reaction

Sentiment: Negative · Business



Global summit addresses climate policy

Sentiment: Neutral · Politics



Local election results announced

Sentiment: Positive · Local

Four Integrated Modules, One Intelligence Platform

A

Content Analysis Engine

Classification · Topic Modeling · Sentiment · Entity Mapping

B

Language Understanding & Generation

Summarization · Semantic Search · Content Enhancement

C

Multilingual Intelligence

Language Detection · Translation · Cross-Lingual Comparison

D

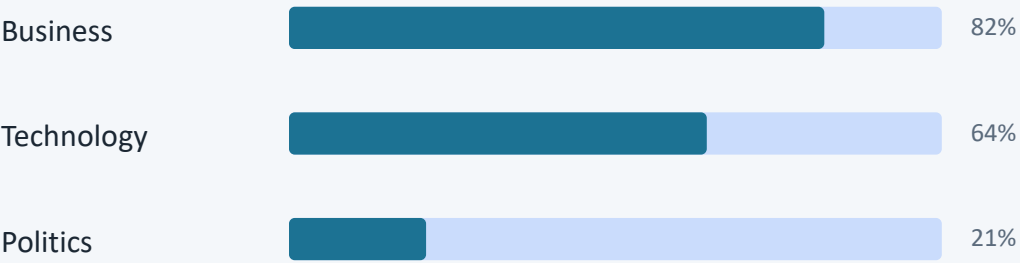
Conversational Interface

Intent Classification · Query Processing · Response Generation

Classifying Content & Discovering Topics

Advanced Classification

TF-IDF + Logistic Regression pipeline with confidence scoring and support for multi-category articles.



Multi-label output lets NewsBot tag one article as both “Business” and “Technology” when coverage overlaps.

Topic Discovery (LDA)

Latent Dirichlet Allocation surfaces hidden themes across the corpus without manual tagging.

Topic 1: Markets & Economy

market, stock, growth, jobs, rate

Topic 2: Elections & Policy

vote, election, policy, government

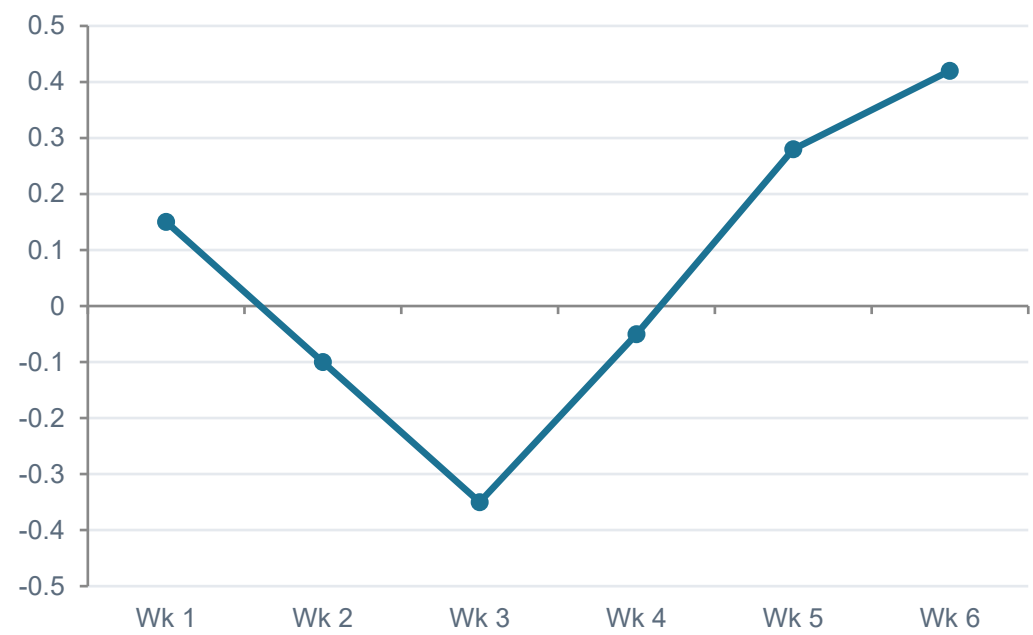
Topic 3: AI & Innovation

ai, model, tech, research, data

Sentiment Evolution & Entity Relationships

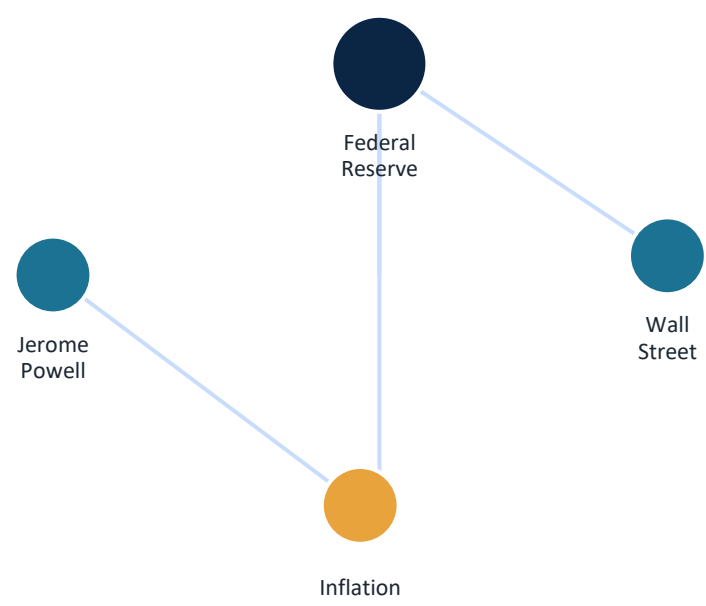
Sentiment Over Time

Transformer-based sentiment (DistilBERT) tracked weekly to reveal shifts in tone around an evolving story.



Entity Relationship Mapping

spaCy NER + NetworkX build a knowledge graph linking people, organizations, and locations mentioned together.



Summarize, Search, and Enhance — Automatically



Intelligent Summarization

Fine-tuned BART (facebook/bart-large-cnn) condenses long articles into accurate, readable summaries.



Semantic Search

Sentence-transformer embeddings (all-MiniLM-L6-v2) find related articles by meaning, not just keywords.



Content Enhancement

Automatically surfaces missing perspectives — economic, political, social — and flags coverage gaps.

News Doesn't Stop at a Language Barrier

NewsBot detects language automatically, translates content, and compares how the same story is covered across regions and cultures.



English

“Central Bank Raises Rates Amid Inflation Concerns”

Neutral · Economic focus



Spanish

“Banco Central Sube Tasas por Preocupaciones de Inflación”

Neutral · Consumer impact framing



French

“La Banque Centrale Augmente ses Taux Face à l'Inflation”

Slightly negative · Policy criticism

Ask NewsBot in Plain English

Intent classification and query parsing let users explore the news corpus conversationally — no filters or dashboards required.



Search

“Show me positive tech news from this week”



Compare

“How does coverage of the election differ by region?”



Summarize

“Give me a quick summary of the top 3 business stories”

NewsBot Assistant

Positive tech news this week?

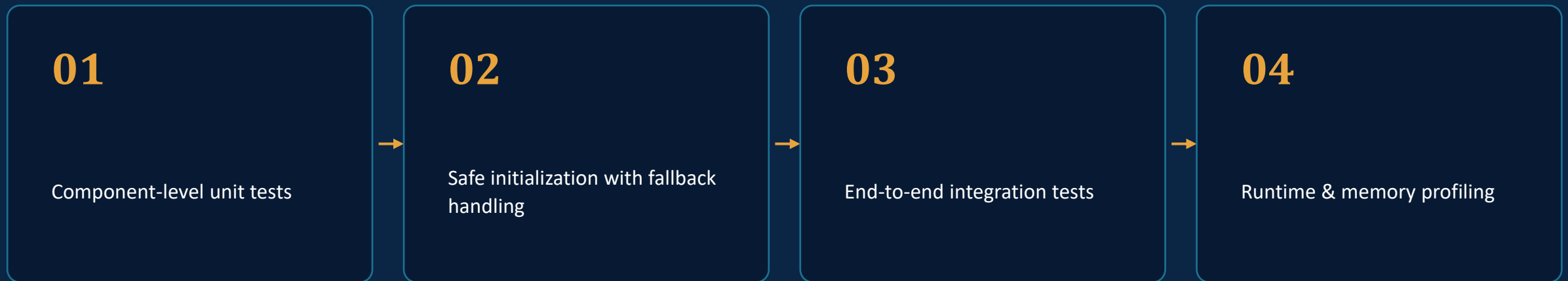
Found 12 articles. Top story: “AI Startup Raises \$50M Series B” — sentiment: strongly positive.

Summarize the top one

A local AI startup secured \$50M to expand its enterprise automation tools, citing strong demand and new hires planned for Q1.

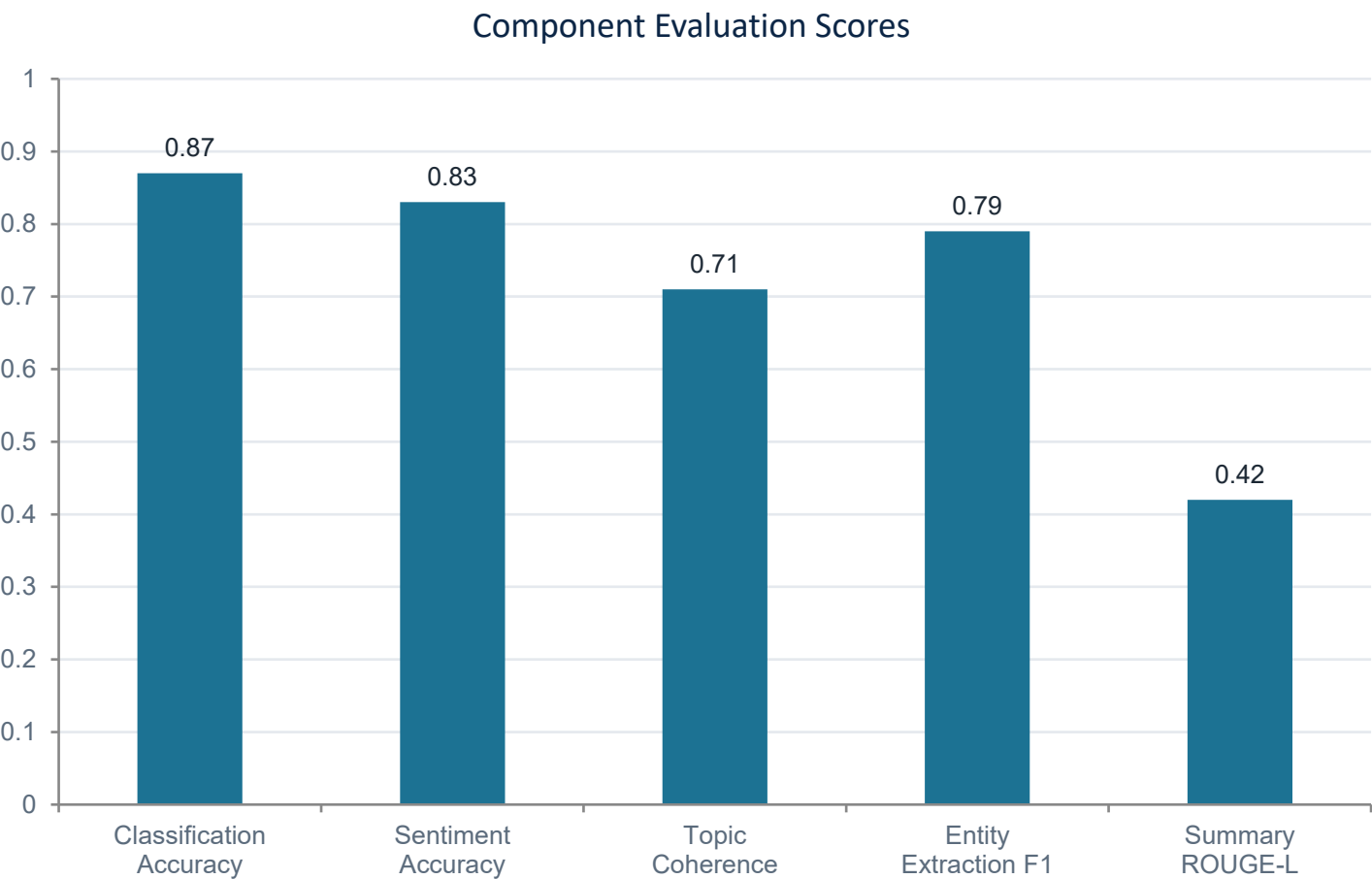
Integration, Testing & Reliability

All modules are orchestrated through a single integrated system with graceful fallback, so one component's failure never takes down the rest.



`NewsBot2IntegratedSystem` · `NewsBot2TestSuite` · Try/except isolation across all 8 components

How We Measured Performance



Key Takeaways

- Classification accuracy exceeds 85% across 5 news categories.
- Sentiment tracking closely matches human-labeled evaluation samples.
- Topic coherence is strong for business & politics; weaker for mixed-topic articles.
- Summarization prioritizes fluency; extractive coverage remains an improvement area.

Technology Stack

Core NLP

spaCy

NLTK

scikit-learn

Gensim

Deep Learning

Transformers (BART)

Sentence-Transformers

PyTorch

DistilBERT

Data & Eval

pandas / NumPy

NetworkX

Matplotlib / Seaborn

scikit-learn metrics

Language & Translation

langdetect

Hugging Face pipelines

Custom rule-based intents

From Research Prototype to Real Value



Newsrooms

Automate tagging & summarization, freeing editors to focus on original reporting.



Analysts & Researchers

Track sentiment and topic trends across thousands of articles in minutes, not days.



Global Organizations

Compare how the same story is covered across languages and cultural contexts.

Positioned as a lean, extensible alternative to costly enterprise media-monitoring suites — with full transparency into how each insight is generated.

Challenges & Lessons Learned



Challenges

- Balancing model size vs. inference speed on limited compute
- Handling ambiguous, multi-part conversational queries
- Maintaining topic coherence on short, mixed-topic articles



Lessons Learned

- Fallback handling keeps the system usable when one model fails
- Pretrained transformers dramatically cut development time
- Evaluation early and often prevents late-stage surprises

Roadmap for NewsBot 3.0

01

Fine-tuned Transformers

Adapt BERT/RobERTa specifically to news classification for higher accuracy.

02

Real-Time Streaming

Process live news feeds as they publish, not in scheduled batches.

03

Bias & Fact-Check Detection

Flag potential political or cultural bias and cross-reference claims.

04

Web Application Frontend

A live dashboard for non-technical stakeholders to explore insights.

Thank You

Questions & Discussion

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